Photovoltaic Series

Planar Diffused Silicon Photodiodes

The Photovoltaic Detector series is utilized for applications requiring high sensitivity and moderate response speeds, with an additional sensitivity in the visible-blue region for the blue enhanced series. The spectral response ranges from 350 to 1100 nm, making the regular photovoltaic devices ideal for visible and near IR applications. For additional sensitivity in the 350 nm to 550 nm region, the blue enhanced devices are more suitable.

These detectors have high shunt resistance and low noise, and exhibit long term stability. Unbiased operation of these detectors offers stability under wide temperature variations in DC or low speed applications. For high light levels (greater than 10mW/cm2), the Photoconductive Series detectors should be considered for better linearity.

These detectors are not designed to be reverse biased! Very slight improvement in response time may be obtained with a slight bias. Applying a reverse bias of more than a few volts (>3V) will permanently damage the detectors. If faster response times are required, the Photoconductive Series should be considered.

Refer to the Photovoltaic Mode (PV) paragraph in the “Photodiode Characteristics” section of this catalog for detailed information on electronics set up.



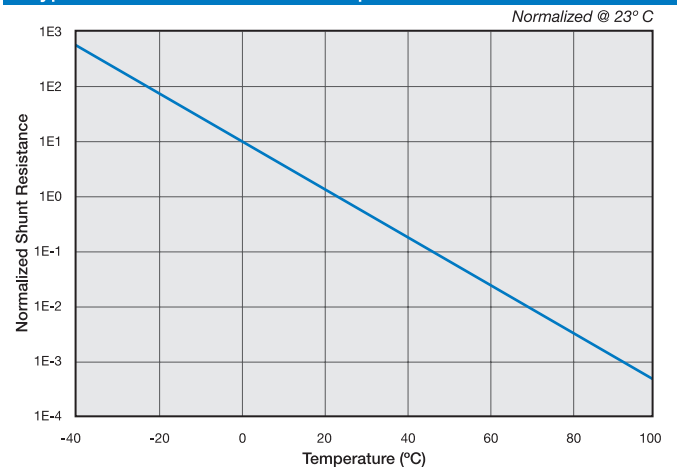
APPLICATIONS

- Colorimeters
- Photometers
- Spectroscopy Equipment
- Fluorescence

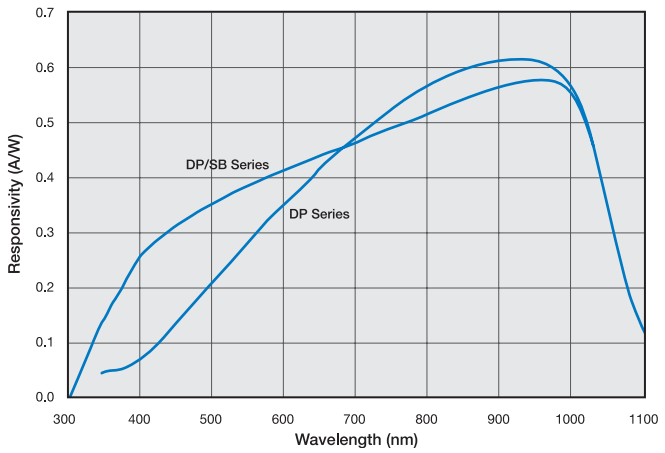
FEATURES

- Ultra Low Noise
- High Shunt Resistance
- Wide Dynamic Range
- Blue Enhanced

Typical Shunt Resistance vs. Temperature



Typical Spectral Response



Photovoltaic Series

Typical Electro-Optical Specifications at T_A=23°C

Model Number	Active Area		Peak Responsivity Wavelength	Responsivity at λp		Capacitance (pF)	Shunt Resistance (GΩ)		NEP (W/√Hz)	Rise Time (ns)	Temp.* Range (°C)		Package Style ¶
	Area (mm²)	Dimensions (mm)	λp (nm)	(A/W)		0 V	-10 mV		0V 970 nm	0 V 632 nm 50 Ω	Operating	Storage	
			typ.	min.	typ.	max.	min.	typ.	typ.	typ.			
'DP' Series, Metal Package													
CD-1705	0.88	0.93 sq	850	0.55	0.60	70	1.0	10	2.1 e-15	2000	-40 ~ +100	-55 ~ +125	71 / Plastic
PIN-2DPI ‡	1.1	0.81 x 1.37	970			150				30			4 / TO-18
PIN-125DPL	1.6	1.27 sq.				160							8 / TO-18
PIN-3CDPI	3.2	1.27 x 2.54				320	0.5	5.0	3.0 e-15	50			4 / TO-18
PIN-3CDP													7 / TO-18
PIN-5DPI	5.1	2.54 ϕ				500	0.4	4.0	3.4 e-15	60			2 / TO-5
PIN-5DP													5 / TO-5
PIN-13DPI	13	3.6 sq				1200	0.35	3.5	3.6 e-15	150			2 / TO-5
PIN-13DP													5 / TO-5
PIN-6DPI	16.4	4.57 ϕ				2000	0.2	2.0	3.9 e-15	220			3 / TO-8
PIN-6DP													6 / TO-8
PIN-44DPI	44	6.6 sq				4300	0.1	1.0	4.8 e-15	475			3 / TO-8
PIN-44DP													6 / TO-8
PIN-10DPI	100	11.28 ϕ				9800	0.05	0.2	6.8 e-15	1000	-10 ~ +60	-20 ~ +70	10/ Lo-Prof
PIN-10DP													11 / BNC
PIN-25DP	613	27.9 ϕ				60000	0.002	0.1	3.0 e-14	6600			12 / BNC
'DP' Series, Plastic Package §													
PIN-220DP	200	10 x 20	970	0.55	0.60	20000	0.02	0.2	1.2 e-14	2200	-10 ~ +60	-20 ~ +70	27 / Plastic

Super Blue Enhanced 'DP/SB' Series, (All Specifications @ λ= 410 nm. V_{BIAS}= 0V, R_L= 50Ω)

Model No.	Active Area/Dimensions		Responsivity (A/W)		Capacitance (pF)	R_{sh} (MΩ)	NEP (W/√Hz)	Operating Current (mA)	Rise Time (µs)			Package Style ¶
	mm²	mm	min.	typ.	typ.	min.	typ.	max.	typ.			
PIN-040DP/SB	0.81	1.02 ϕ	0.15	0.20	60	600	2.0 e-14	0.5	0.02	-10 ~ +60	-20 ~ +70	1 / TO-18
PIN-5DP/SB †	5.1	2.54 ϕ			450	150	5.2 e-14	2.0	0.2			5 / TO-5
PIN-10DP/SB	100	11.28 ϕ			8800	10	2.0 e-13	10.0	2.0			11 / BNC
PIN-10DPI/SB												10 / Metal
PIN-220DP/SB	200	10 x 20			17000	5	2.9 e-13	10.0	4.0			27 / Plastic

'5T' Series, Blue

Model No.	Active Area/Dimensions		Responsivity (A/W) 436nm		Capacitance (pF) 0V	R_{sh} (MΩ)	NEP (W/√Hz)	Dark Current (pA)	Rise Time (µs)			Package Style ¶
	mm²	mm	min.	typ.	max	min.	typ.	max.	typ.			
OSD1-5T	1.0	1.0 sq	0.18	0.21	35	250	2.5 e-14	1.0	7	-25 ~ +75	-45 ~ +100	7 / TO-18
OSD3-5T	3.0	2.5 x 1.2			80	100	3.0 e-14	2.0	9			7 / TO-18
OSD5-5T	5.0	2.5 ϕ			130	100	3.3 e-14	2.0	9			5 / TO-5
OSD15-5T	15.0	3.8 sq			390	50	5.6 e-14	10.0	12			5 / TO-5
OSD60-5T	62.0	7.9 sq			1800	3	2.1 e-13	25.0	30			72 / TO-8
OSD100-5TA	100.0	11.3 ϕ			2500	2	2.5 e-13	30.0	45			74 / Special

‡ The 'I' suffix on the model number is indicative of the photodiode chip being isolated from the package by an additional pin connected to the case.
§ The photodiode chips in "FIL" series are isolated in a low profile plastic package. The have a large field of view as well as "in line" pins.
¶ For mechanical drawings please refer to pages 58 thru 69.
† Operating Temperature: -40 to +100 °C, Storage Temperature: -55 to +125 °C.
* Non-Condensing temperature and Storage Range, Non-Condensing Environment.

OSI Optoelectronics An OSI Systems Company +1 310-978-0516 | www.osioptoelectronics.com

Mechanical Drawings

Mechanical Specifications and Die Topography

1. Parameter Definitions:

A = Distance from top of chip to top of glass.
a = Photodiode Anode.
B = Distance from top of glass to bottom of case.
c = Photodiode Cathode
(Note: cathode is common to case in metal package products unless otherwise noted).
W = Window Diameter.
F.O.V. = Filed of View (see definition below).

2. Dimensions are in inches (1 inch = 25.4 mm).

3. Pin diameters are 0.018 ± 0.002" unless otherwise specified.

4. Tolerances (unless otherwise noted)

General: 0.XX ±0.01"
0.XXX ±0.005"
Chip Centering: ±0.010"
Dimension 'A': ±0.015"

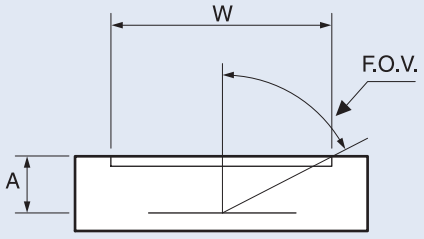
5. Windows

All **'UV'** Enhanced products are provided with QUARTZ glass windows, 0.027 ± 0.002" thick.

All **'XUV'** products are provided with removable windows.

All **'DLS'** PSD products are provided with A/R coated glass windows.

All **'FIL'** photoconductive and photovoltaic products are epoxy filled instead of glass windows.



$$F.O.V. = \tan^{-1}\left(\frac{W}{2A}\right)$$



For Further Assistance
Please Call One of Our Experienced
Sales and Applications Engineers

310-978-0516



- Or -
On the Internet at

www.osioptoelectronics.com

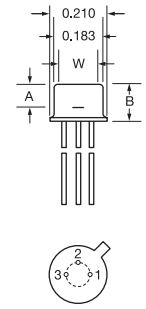
Mechanical Specifications

All units in inches. Pinouts are bottom view.

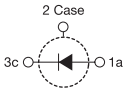
1 TO-18

Products:

PIN-020A
PIN-040A
PIN-040-DP/SB



Pin Circle Dia.=0.100

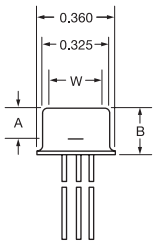


P/N	A	B	W
PIN-020A	0.075	0.200	0.155
PIN-040A	0.075	0.200	0.155

2 TO-5

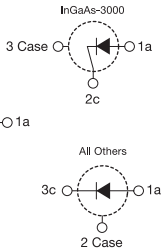
Products:

PIN-5DI
PIN-5DPI
PIN-13DI
PIN-13DPI
PIN-5-YAG
CD-25T





Pin Circle Dia.=0.200

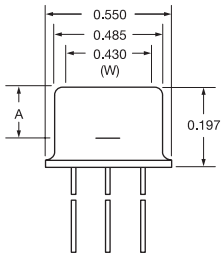


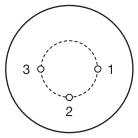
P/N	A	B	W
All Others	0.094	0.180	0.240
CD-25T	0.050	0.130	0.23

3 TO-8

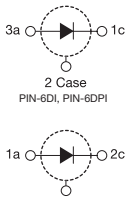
Products:

PIN-6DI
PIN-6DPI
PIN-44DI
PIN-44DPI
OSD35-0
OSD35-7Q





Pin Circle Dia.=0.295

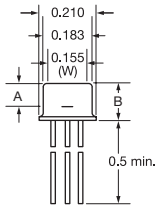


P/N	A
PIN-6DI/6DPI	0.115
PIN-44DI/44DPI	0.125
OSD35-0	0.130
OSD35-7Q	0.130 Quartz Window

4 TO-18

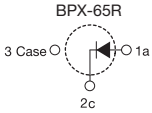
Products:

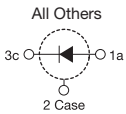
PIN-2DI
PIN-2DPI
PIN-3CDP
PIN-3CDPI
BPX-65R





Pin Circle Dia.=0.100



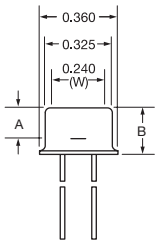


P/N	A	B
BPX-65R	0.070	0.200
All Others	0.090	0.150

5 TO-5

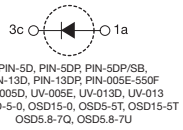
Products:

PIN-5D
PIN-5DP
PIN-5DP/SB
PIN-13D
PIN-13DP
PIN-005E-550F
UV-001
UV-005
UV-005D
UV-005E
UV-013D
UV-013E
UV-015
OSD-5-0
OSD15-0
OSD5-5T
OSD15-5T
OSD5.8-7Q
OSD5.8-7U

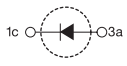




Pin Circle Dia.=0.200



PIN-5D, PIN-5DP, PIN-5DP/SB,
PIN-13D, PIN-13DP, PIN-005E-550F
UV-005D, UV-005E, UV-013D, UV-013
15OSD-5-0, OSD15-0, OSD5-5T, OSD15-5T
OSD5.8-7Q, OSD5.8-7U



UV-001, UV-005, UV-015

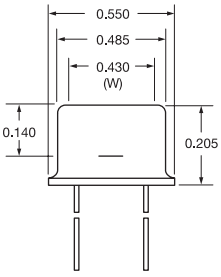
P/N	A	B
OSD-Prefix Devices	0.050	0.130
All Others	0.102	0.180

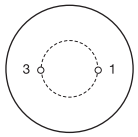
Quartz Window: OSD5.8-7Q
UV Transmissive Window: OSD5.8-7U

6 TO-8

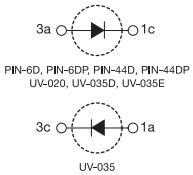
Products:

PIN-6D
PIN-6DP
PIN-44D
PIN-44DP
UV-020
UV-035D
UV-035E
UV-035





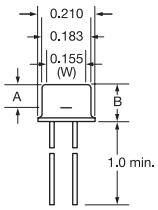
Pin Circle Dia.=0.295



7 TO-18

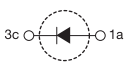
Products:

PIN-3CD
PIN-3CDP
BPX-65
OSD1-0
OSD1-5T
OSD3-5T
OSD1.2-7Q
OSD1.2-7U





Pin Circle Dia.=0.100



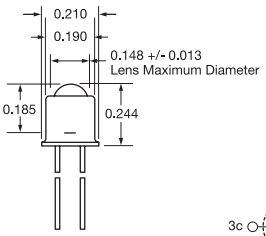
P/N	A	B
PIN-3CD / 3CDP	0.087	0.146
BPX-65	0.075	0.200
OSD-Prefix Devices	0.080	0.200

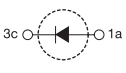
Quartz Window: OSD1.2-7Q
UV Transmissive Window: OSD1.2-7U

8 TO-18

Products:

PIN-125DPL





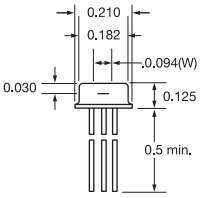


Pin Circle Dia.=0.100

9 TO-18

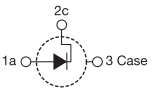
Products:

PIN-HR005
PIN-HR008
PIN-HR020
PIN-HR026
PIN-HR040





Pin Circle Dia.=0.100

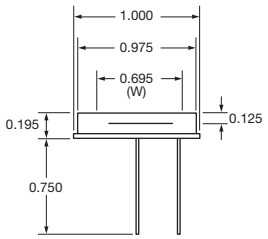


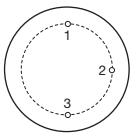
All units in inches. Pinouts are bottom view.

10 Low Profile

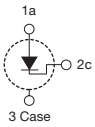
Products:

PIN-10DI
PIN-10DPI
PIN-10DPI/SB
UV-50L
UV-100L





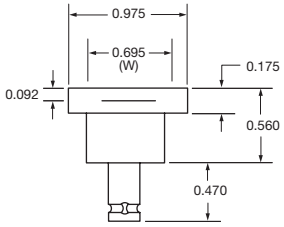
Pin Circle Dia.=0.73



11 BNC

Products:

PIN-10D
PIN-10DP
PIN-10DP/SB
UV-50
UV-100

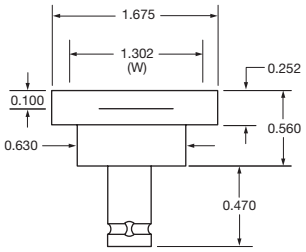


Outer Contact — Anode	PIN-10D, PIN-10DP, PIN-10DP/SB
Outer Contact — Cathode	UV-50, UV-100

12 BNC

Products:

PIN-25D
PIN-25DP

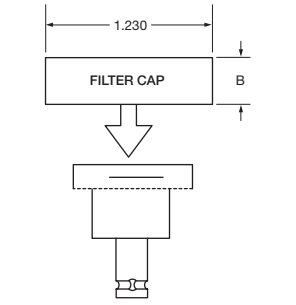


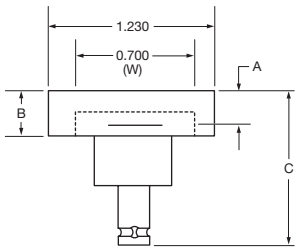
Outer Contact — Anode

13 Special BNC

Products:

PIN-10AP
PIN-10DF





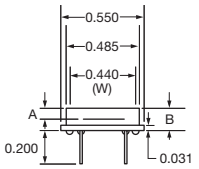
Dimensions

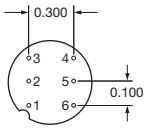
P/N	A	B	C
PIN-10DF	0.217	0.330	1.020
PIN-10AP	0.386	0.550	1.415

14 Special Plastic

Products:

FIL-5C
FIL-20C
FIL-UV20
FIL-C4DG





Dimensions

P/N	A	B
FIL-5C FIL-20C	0.060	0.130
FIL-UV005 FIL-UV20 FIL-C4DG	0.087	0.152

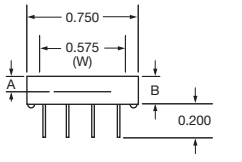
Pinouts

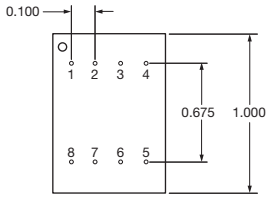
P/N	1	2	3	4	5	6
FIL-5C FIL-20C FIL-UV005	a	-	c	a	-	c
FIL-UV20	c	-	a	c	-	a
FIL-C4DG	c	a	c	c	a	c

15 Special Plastic

Products:

FIL-44C
FIL-100C
FIL-UV50
FIL-UV100
FIL-C10DG





Dimensions

P/N	A	B
FIL-44C FIL-100C	0.052	0.130
FIL-UV50 FIL-UV100	0.090	0.155
FIL-C10DG	0.082	0.155

Pinouts

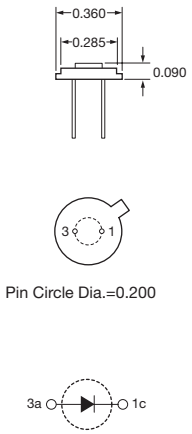
P/N	1	2	3	4	5	6	7	8
FIL-44C FIL-100C	a	-	-	c	a	-	-	c
FIL-UV50 FIL-UV100	c	-	-	a	c	-	-	a
FIL-C10DG	c	a	a	c	c	a	a	c

All units in inches. Pinouts are bottom view.

22 TO-5

Products:

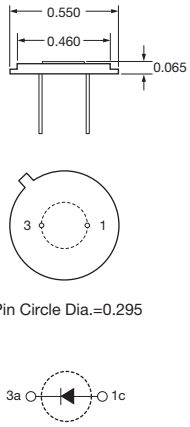
XUV-005



23 TO-8

Products:

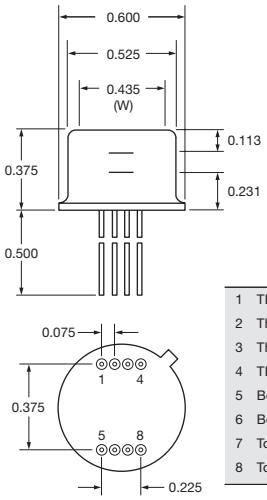
XUV-020
XUV-035



24 TO-8

Products:

PIN-DSIn-TEC



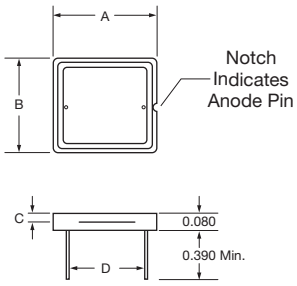
Pinout

1	TEC (-)
2	Thermistor
3	Thermistor
4	TEC (+)
5	Bottom InGaAs, Cathode
6	Bottom InGaAs, Anode
7	Top Silicon, Anode
8	Top Silicon, Cathode

25 Special Ceramic / Plastic

Products:

RD-100
RD-100A
UV-35P
UV-005EC
UV-035EC
UV-100EC
UV-005DC
UV-035DC
UV-100DC
XUV-50C
XUV-100C
OSD35-7CO
OSD35-LR-A
OSD35-LR-D



Dimensions

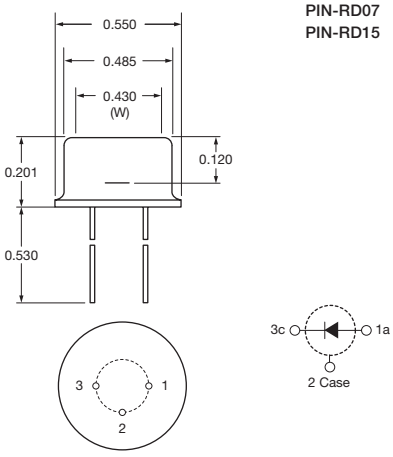
P/N	A	B	C	D
UV-005EC	0.400	0.350	0.030	0.280
UV-035EC	0.400	0.350	0.030	0.290
UV-100EC	0.650	0.590	0.048	0.500
UV-005DC	0.400	0.350	0.030	0.280
UV-035DC	0.400	0.350	0.030	0.290
UV-100DC	0.650	0.590	0.053	0.500
XUV-50C	0.650	0.590	0.027	0.490
XUV-100C	0.650	0.590	0.027	0.490
RD-100	0.650	0.590	0.027	0.490
RD-100A	0.650	0.590	0.027	0.490
UV-35P	0.390	0.345	0.050	0.275
OSD35-7CO	0.390	0.350	---	0.290
OSD35-LR-A	0.390	0.350	---	0.290
OSD35-LR-D	0.390	0.350	---	0.290

Note: OSD35-prefix packages come with 0.31" (min.) leads

26 TO-8

Products:

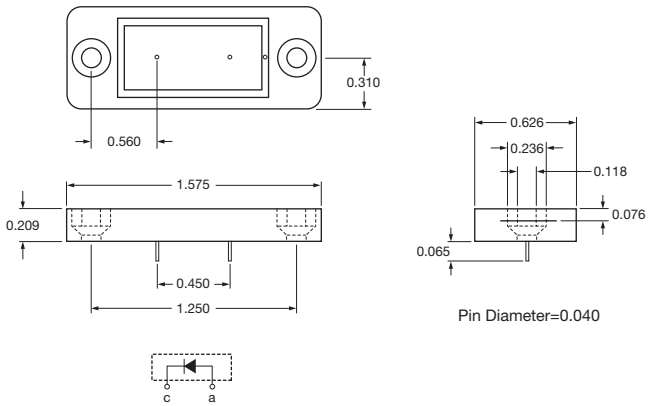
PIN-RD07
PIN-RD15



27 Special Plastic

Products:

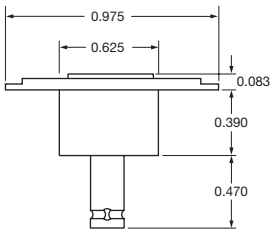
PIN-220D
PIN-220DP
PIN-220DP/SB



28 BNC

Products:

XUV-100



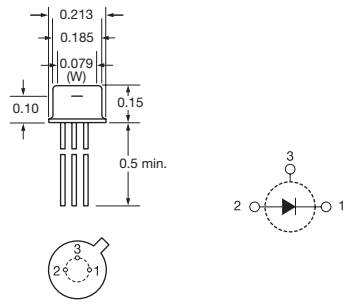
BNC Connector
Outer Contact = Cathode

All units in inches. Pinouts are bottom view.

68 TO-18

Products:

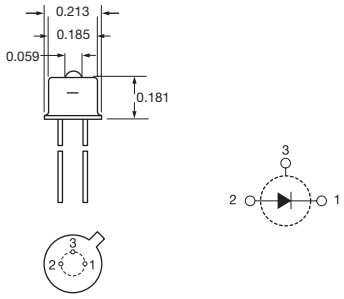
APD-300
APD-500



69 TO-18

Products:

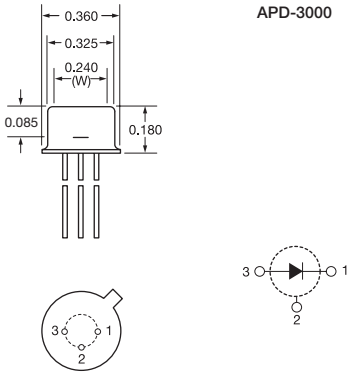
APD-300L
APD-500L



70 TO-5

Products:

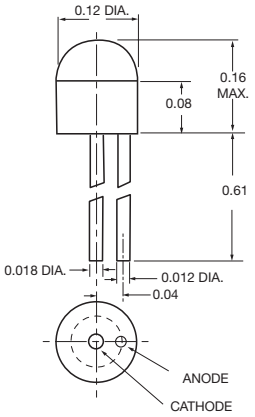
APD-900
APD-1500
APD-3000



71 Plastic

Products:

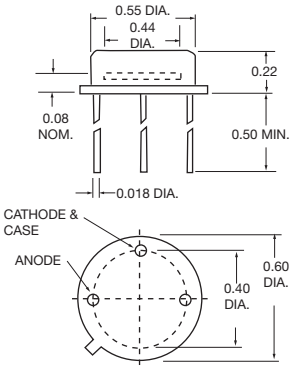
CD-1705



72 TO-8

Products:

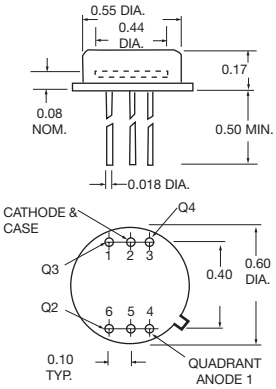
OSD-60-0



73 TO-8

Products:

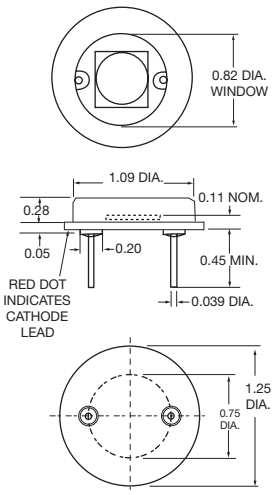
QD50-0



74 Special

Products:

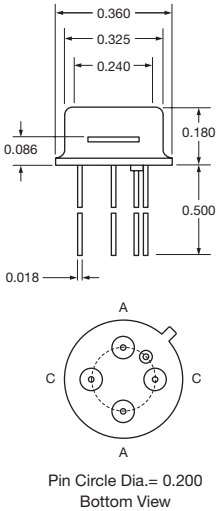
OSD100-0A
OSD100-5TA



75 TO-5

Products:

DLS-2S



76 Plastic Molded

Products:

OS-P200

